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Scanning assembly and ultrasound imaging device

Abstract

A scanning assembly, including: a housing, wherein a lower portion of the housing comprises an opening, and the opening can be closed by tissue to be scanned; a pressure difference generating device, disposed on the housing and used to generate an air pressure difference between the interior and the exterior of the housing, the air pressure difference providing a first pressure for the tissue to be scanned, the weight of the scanning assembly providing a second pressure for the tissue to be scanned, the tissue to be scanned being compressed under the action of a compressive force, and the compressive force comprising the first pressure and the second pressure; and an ultrasonic transducer, disposed in the housing and used to perform ultrasound scanning on the compressed tissue to be scanned.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) The present application claims priority to Chinese Patent Application No. 202310187979.7, filed on Feb. 28, 2023. The entire contents of the above-listed application are incorporated by reference herein in their entirety.

TECHNICAL FIELD

(2) The present application relates to the field of medical imaging, and relates in particular to a scanning assembly and an ultrasound imaging device.
BACKGROUND
(3) Ultrasound imaging is a non-destructive and real-time imaging means, which can be used for scanning a variety of human organs and tissues. One of the examples is an automatic breast ultrasound imaging system which can perform ultrasound imaging on the breast of a person receiving a scan. In some examples, the automatic breast ultrasound imaging system includes a scanning assembly capable of scanning breasts. In order to perform high-quality imaging on breasts, it is generally necessary to apply an appropriate downward compressive force to the breasts. In order to ensure the safety of a person receiving a scan, it is further necessary to ensure that the compressive force is not set to be overly strong. Existing solutions include providing a balancing mechanism that includes a counterweight, and connecting the counterweight to a scanning assembly by means of an arm. The counterweight can balance the gravity of the scanning assembly to adjust, under the action of a driving mechanism, the downward pressure of the scanning assembly.
(4) The above-described method for adjusting the compressive force solves, to some extent, the problem of how to reasonably adjust the compressive force on breasts, but still has notable defects. In one aspect, configurations of the counterweight and the arm obviously affect the range of movement of the scanning assembly, and during use, a scanning operator has to reasonably position the ultrasound imaging device and a person receiving a scan. In another aspect, providing structures such as the counterweight and the arm increases the complexity of the imaging device, thereby increasing the complexity of operations performed by a scanning operator.
SUMMARY
(5) The aforementioned defects, deficiencies, and problems are solved herein, and these problems and solutions will be understood through reading and understanding the following description.
(6) Provided in some embodiments of the present application is a scanning assembly, comprising: a housing, wherein a lower portion of the housing comprises an opening, and the opening can be closed by tissue to be scanned; a pressure difference generating device, disposed on the housing and used to generate an air pressure difference between the interior and the exterior of the housing, the air pressure difference providing a first pressure for the tissue to be scanned, the weight of the scanning assembly providing a second pressure for the tissue to be scanned, the tissue to be scanned being compressed under the action of a compressive force, and the compressive force comprising the first pressure and the second pressure; and an ultrasonic transducer, disposed in the housing and used to perform ultrasound scanning on the compressed tissue to be scanned. The present application also provides an ultrasound imaging device comprising the above-described scanning assembly.
(7) Further provided in some other embodiments of the present application is an ultrasound imaging device, comprising the scanning assembly as described above.
(8) It should be understood that the brief description above is provided to introduce, in a simplified form, concepts that will be further described in the detailed description. The brief description above is not meant to identify key or essential features of the claimed subject matter. The scope is defined uniquely by the claims that follow the detailed description. Furthermore, the claimed subject matter is not limited to implementations that solve any deficiencies raised above or in any section of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The present application will be better understood by reading the following description of non-

limiting embodiments with reference to the accompanying drawings, wherein:
(2) FIG. **1** shows a perspective view of an ultrasound imaging device in the prior art;
(3) FIG. **2** shows a schematic diagram of a scanning assembly according to some embodiments of the present application;
(4) FIG. **3** shows a schematic diagram of a scanning assembly from which part of the top structure has been removed according to some embodiments of the present application;
(5) FIG. **4** shows a perspective view of a scanning assembly from the bottom according to some embodiments of the present application; and
(6) FIG. **5** shows a schematic diagram of an ultrasound imaging device comprising a scanning assembly according to some embodiments of the present application.

DETAILED DESCRIPTION

(7) Specific embodiments of the present application are described below. It should be noted that in the specific description of said embodiments, for the sake of brevity and conciseness, the present description cannot describe all of the features of the actual embodiments in detail. It should be understood that in the actual implementation process of any embodiment, just as in the process of any one engineering project or design project, a variety of specific decisions are often made to achieve specific goals of the developer and to meet system-related or business-related constraints, which may also vary from one embodiment to another. Furthermore, it should also be understood that although efforts made in such development processes may be complex and extended, for a person of ordinary skill in the art related to the disclosure of the present application, some design, manufacture or production changes made on the basis of the technical disclosure of the present disclosure are only conventional technical means, and should not be construed that the content of the present disclosure is insufficient.

(8) Unless otherwise defined, the technical or scientific terms used in the claims and the description should be as they are usually understood by those possessing ordinary skill in the technical field to which they belong. Terms such as “first”, “second” and similar terms used in the present description and claims do not denote any order, quantity, or importance, but are only intended to distinguish different constituents. The terms “one” or “a/an” and similar terms do not express a limitation of quantity, but rather that at least one is present. The terms “include” or “comprise” and similar words indicate that an element or object preceding the terms “include” or “comprise” encompasses elements or objects and equivalent elements thereof listed after the terms “include” or “comprise”, and do not exclude other elements or objects. The terms “connect” or “link” and similar words are not limited to physical or mechanical connections, and are not limited to direct or indirect connections.

(9) Although some embodiments of the present application are presented in a particular context of human breast ultrasound, it should be understood that the present application is applicable to performing an ultrasound scan on any externally accessible human or animal body part (for example, the abdomen, legs, feet, arms, or neck).

(10) FIG. **1** shows a perspective view of an ultrasound imaging device **102** in the prior art. The body of the ultrasound imaging device **102** may include a main machine, a display **110**, an adjustable arm **106**, and a scanning assembly **108**. The adjustable arm **106** and an internal balancing mechanism (e.g., a counterweight and a related connecting member) thereof can be used to balance the weight of the scanning assembly **108** and adjust, during ultrasound scanning, a compressive force applied by the scanning assembly **108** onto a person receiving a scan. The ultrasound imaging device **102** is exemplarily described below.

(11) The ultrasound imaging device **102** may include a body frame **104**, an ultrasonic processor housing **105** including an ultrasonic processor, a movable and adjustable support arm (for example, an adjustable arm) **106** including a hinge joint **114**, a scanning assembly **108** connected to a first end **120** of the adjustable arm **106** by means of a ball-and-socket connector (for example, a ball joint) **112**, and a display **110** connected to the body frame **104**. The display **110** is connected to the

body frame **104** at a joining point where the adjustable arm **106** enters the body frame **104**. Since the display **110** is directly connected to the body frame **104** rather than the adjustable arm **106**, the display **110** does not affect the weight of the adjustable arm **106** and a balancing mechanism of the adjustable arm **106**.

(12) The imaging device **102** shown in FIG. **1** allows a user to adjust the compressive force of the scanning assembly **108** on tissue to be scanned. An exemplary description is provided below. In one embodiment, the adjustable arm **106** is configured so that the scanning assembly **108** is substantially neutrally buoyant in a space, and in this case, the scanning assembly **108** can be operated to hover in any position within a certain height range. In another example, the adjustable arm **106** is configured so that the scanning assembly **108** has a light net downward weight (for example, 1-2 kg) for pressing down on the breasts, while allowing easy user operation. The scanning assembly **108** that is neutrally buoyant or has a light downward weight facilitates the movement of the scanning assembly **108** performed by the user to position the same on a surface of the tissue to be scanned. After the scanning assembly **108** is positioned, internal components of the ultrasound imaging device **102** may be adjusted to apply a desired downward weight for pressing down on the breast and improved image quality. In one example, the internal components may include a counterweight (not shown in the drawing). By controlling a balancing force of the counterweight on the scanning assembly **108**, the user can control the compressive force of the scanning assembly **108** on the tissue to be scanned. In another example, the internal components may further include a motor and a transmission thereof (not shown in the drawing). The motor and the transmission thereof may be used to control the adjustment of the balancing force of the counterweight on the scanning assembly **108**.

(13) Further, the scanning assembly **108** may include a film **118** that is in a substantially tensioned state and that is at least partially attached, said film being used for pressing down on the breast. The film **118** has a bottom surface for making contact with the breast, and when the bottom surface makes contact with the breast, the transducer sweeps over a top surface of the film to scan the breast. In one example, the film is **118** a tensioned fabric sheet. In another example, the film **118** is a hard plastic sheet. The compressive force of the scanning assembly **108** can act on the tissue to be scanned by means of the film **118**.

(14) In order to improve the flexibility of movement of the scanning assembly **108**, the ultrasound imaging device **102** is further configured to have an additional structure. In addition to the adjustable arm **106** and the counterweight that are exemplarily described above and that are adjustable in the vertical height direction, the adjustable arm **106** includes a hinge joint **114** to allow the scanning assembly **108** to move in the horizontal direction. The hinge joint **114** divides the adjustable arm **106** into a first arm portion and a second arm portion. The first arm portion is connected to the scanning assembly **108** and the second arm portion is connected to the body frame **104**. The hinge joint **114** allows the second arm portion to rotate relative to the second arm portion and the body frame **104**. For example, the hinge joint **114** allows the scanning assembly **108** to translate transversely and horizontally, but not vertically, relative to the second arm portion and the body frame **104**. In such manner, the scanning assembly **108** can rotate toward the body frame **104** or away from the body frame **104**. However, the hinge joint **114** is configured to allow the entire adjustable arm **106** (for example, the first arm portion and the second arm portion) to move vertically together as a whole (for example, translating upward and downward along with the body frame **104**). By using the above means, the scanning assembly **108** can move vertically and/or horizontally within a certain range to move close to or away from a site to be scanned.

(15) Optionally, the adjustable arm may include a potentiometer (not shown) to allow position and direction sensing performed by the pressing/scanning assembly **108**, or may use other types of position and direction sensing (such as gyroscope, magnetic, optical, and radio frequency (RF)). A fully functional ultrasonic engine may be provided within the ultrasonic processor housing **105**, and is configured to drive the ultrasonic transducer, and generate volumetric breast ultrasound data

from a scan in conjunction with related position and orientation information. In some examples, volumetric scan data may be transmitted to another computer system by using any of a variety of data transmission methods known in the art so as to be further processed, or the volumetric scan data may be processed by the ultrasonic engine. A general-purpose computer/processor integrated with the ultrasonic engine may further be provided for general user interface and system control. The general-purpose computer may be a self-contained stand-alone unit, or may be remotely controlled, configured, and/or monitored by remote stations connected across networks.

(16) In the ultrasound imaging device **102** illustrated above, the adjustable arm **106** is connected to the scanning assembly **108** to adjust the compressive force required by the scanning assembly **108** to perform ultrasound scanning, and can ensure that the compressive force is within a reasonable range so as to not put a person receiving a scan in danger. It is not difficult to see that the adjustable arm **106** includes a plurality of components, thereby increasing the complexity of operation performed by the user. Moreover, the scanning assembly **108** is connected to the adjustable arm **106**, so that the range of movement of the scanning assembly **108** is limited by the range of movement of the adjustable arm **106**. For example, in the horizontal direction, the scanning assembly **108** can move within a certain range around the hinge joint **114**, and in the vertical direction, the scanning assembly **108** is restricted by the vertical travel of the adjustable arm **106**.

(17) In view of this, improvements are provided in embodiments of the present application in the hope of maintaining as much as possible the flexibility of scanning performed by a scanning assembly while ensuring that the compressive force of the scanning assembly on tissue to be scanned is configured to be within a reasonable and safe range.

(18) Referring to FIG. **2**, FIG. **2** shows a schematic diagram of a scanning assembly **200** according to some embodiments of the present application. The scanning assembly **200** is adapted to perform ultrasound scanning on a supine person receiving a scan. The scanning assembly **200** may include a housing **201**, a pressure difference generating device **202**, and an ultrasonic transducer **203**.

(19) A lower portion of the housing **201** may include an opening **211**. The opening **211** can be closed by tissue to be scanned (not shown in FIG. **2**).

(20) The pressure difference generating device **202** is disposed on the housing **201**, and is used to generate an air pressure difference between the interior and the exterior of the housing **201**. The air pressure difference provides a first pressure for the tissue to be scanned, and the weight of the scanning assembly **200** provides a second pressure for the tissue to be scanned. The tissue to be scanned is compressed under the action of a compressive force, and the compressive force includes the first pressure and the second pressure.

(21) The ultrasonic transducer **203** is disposed in the housing **201**, and is used to perform ultrasound scanning on the compressed tissue to be scanned.

(22) In such a configuration means, part of the compressive force on the tissue to be scanned is provided by the weight of the scanning assembly **200**, and part is provided by the pressure difference generating device **202**, so that no other additional assembly is further needed. The scanning assembly **200** can be moved with a high degree of freedom without being overly restricted by any additional structures (e.g., an adjustable arm or other assemblies). In addition, the pressure difference generating device **202** is disposed on the housing **201**, i.e., as part of the entirety of the scanning assembly **200**, and likewise does not need any additional connecting structures, and therefore the pressure difference generating device **202** likewise does not affect the flexibility of use of the scanning assembly **200**.

(23) In addition, the compressive force of the scanning assembly **200** of the present application on the tissue to be scanned at least includes two parts. One part is from the weight of the scanning assembly **200** (the second pressure), and the other part is from the pressure difference (the first pressure) generated by the pressure difference generating device **202** on the scanning assembly **200**. The first pressure and the second pressure are of different types, so that the scanning assembly **200** of the present application has higher safety and operability. Since the pressure difference

generating device **202** can be used to provide the first pressure as the compressive force, the scanning assembly **200** that uses the weight thereof to provide the second pressure can be configured to be light (e.g., less than 3 kilograms). In this way, when scanning is performed on the person receiving a scan, there is no danger even if the weight of the entire scanning assembly **200** acts on the person receiving a scan. Moreover, since the weight of the scanning assembly **200** acting as the second pressure can provide part of the compressive force, the first pressure provided by the pressure difference generating device **202** does not need to be overly high. In this way, the air pressure difference between the interior and the exterior of the scanning assembly **200** is not so large that the skin surface of the person receiving a scan will be injured due to a negative pressure. It should be noted that in some embodiments, the sum of the first pressure and the second pressure is the compressive force of the scanning assembly **200** on the tissue to be scanned. In some other embodiments, the compressive force may further include another force, e.g., an appropriate downward pressure applied by a scanning operator operating the scanning assembly **200**. Examples are not exhaustively enumerated.

(24) It can be understood that the material and shape of the housing **201** in the above embodiments may be arbitrary. In one embodiment, the housing **201** may include a transparent high-molecular polymer to facilitate the scanning operator to observe, through the housing **201**, a scanned site and scanning condition of a probe. In another embodiment, the housing **201** may further include a metal material to improve the rigidity of the entire scanning assembly **200**. The shape of the housing **201** may be an arbitrary shape, such as a circle, a rectangle, etc. Correspondingly, the shape of the opening **211** of the lower portion of the housing **201** may also be arbitrary, as long as the opening **211** and the tissue to be scanned can cooperate with each other to form a substantially closed space, so as to facilitate the pressure difference generating device **202** to generate the air pressure difference between the interior and the exterior of the scanning assembly **200**. Moreover, the housing **201** may further include different components. For example, as shown in FIG. **2**, the housing **201** may include a surrounding frame **212** and a cover **213** located above. The cover **213** may be made of a transparent material to facilitate the scanning operator to observe the scanning assembly **200**. The frame **212** may have a certain thickness to facilitate integration therein of a cable or another structure. In one example, the cover **213** may be fixedly connected to the frame **212**. In another example, the cover **213** may be detachably/movably connected to the frame **212**. Such a configuration means facilitates the scanning operator to clean and sterilize the interior of the scanning assembly **200** when needed.

(25) With continued reference to FIG. **2**, the pressure difference generating device **202** may be disposed in any position on the housing **201**. For example, the pressure difference generating device **202** may be disposed on the frame **212** of the housing **201**. On the basis that the tissue to be scanned closes the opening **211** to cause a closed space to be formed between the scanning assembly **200** and the tissue to be scanned, the pressure difference generating device **202** can pump air from the closed space out of the space so as to generate an air pressure difference between the interior and the exterior of the closed space. The air pressure difference causes the scanning assembly **200** to press down on the tissue to be scanned. Since in the present application, part of the compressive force on the tissue to be scanned is provided by the weight of the scanning assembly **200**, the pressure difference generating device **202** does not need to generate a pressure difference of very high power. Therefore, as shown in FIG. **2**, the pressure difference generating device **202** can be configured to have smaller power and size, and be integrated on the housing **201** of the scanning assembly **200**. The pressure difference generating device **202** is an independent device disposed on the housing **201**, and does not need any external apparatuses (e.g., a vacuuming device, or the like).

(26) Considering that compressive forces needed by different persons receiving a scan may be different and that even the same person receiving a scan may need different compressive forces depending on different compression conditions, improvements are provided in some embodiments

of the present application. In some embodiments, the pressure difference generating device **202** is configured to be capable of adjusting the magnitude of the air pressure difference, to thereby adjust the first pressure. The above-described adjustment method may be implemented by changing the power of the pressure difference generating device **202**. When the pressure difference generating device **202** operates at higher power, more air is drawn out from the closed space between the scanning assembly **200** and the tissue to be scanned, thereby resulting in a greater air pressure drop. At which time, the air pressure difference is increased. When the pressure difference generating device **202** operates at lower power, less air is drawn out from the closed space. At which time, the air pressure drop is smaller, and thus the air pressure difference is smaller.

(27) In some embodiments, the pressure difference generating device **202** may include a fan. The rotational speed of the fan may be configured to be adjustable, thereby enabling the magnitude of the air pressure difference to be adjusted. As shown in FIG. **2**, the pressure difference generating device **202** including the fan may be configured to communicate with the interior and the exterior of the housing **201** so as to remove air from an internal space of the scanning assembly to the outside to generate a pressure difference.

(28) The fan acting as the pressure difference generating device **202** has advantages in a plurality of aspects. In one aspect, as compared with other negative pressure devices, such as a pump, pumping power of the fan is not overly high and thus does not pose a danger even if an error occurs during use of the scanning assembly **200**. In addition, as described herein above, the scanning assembly **200** of the present application provides part of the compressive force by means of the weight thereof, and therefore the problem in which the compressive force provided by the fan is not sufficient does not occur. In another aspect, air channels inside and outside of the fan are connected, so that once a person receiving a scan feels uncomfortable, only a power-off operation is required to ensure that the pressure difference between the interior and the exterior rapidly returns to a balanced state, thereby rapidly removing the first pressure. The pressure difference generating device **202** of the present application can further ensure the safety of the person receiving a scan.

(29) It can be understood that although FIG. **2** shows only one pressure difference generating device **202**, a plurality of pressure difference generating devices **202** may be provided in other embodiments of the present application. For example, a plurality of pressure difference generating devices **202** may be uniformly disposed on the housing. In one aspect, a plurality of pressure difference generating devices **202** can further increase the first pressure. In another aspect, a plurality of pressure difference generating devices **202** being uniformly disposed can further ensure that the downward pressure between the scanning assembly **200** and the tissue to be scanned is balanced, thereby ensuring that a high-quality ultrasound scanning image can be acquired.

(30) Taking into consideration the faster control of the pressure difference generating device **202**, in some embodiments, the scanning assembly **200** may further include a control unit **204**. The control unit **204** is configured to, when operated, control the pressure difference generating device **202** to adjust the magnitude of the first pressure to thereby adjust the magnitude of the compressive force. The control unit **204** is provided, so that the magnitude of the compressive force can be easily adjusted without any additional complex compressive force adjustment device, and high-quality imaging is facilitated.

(31) In some embodiments, the control unit **204** may include a button. Correspondingly, a user may operate the control unit **204** by pressing or other means. In a non-limiting embodiment, the user can control, by pressing the control unit **204**, the pressure difference generating device **202** to gradually adjust the increase of the first pressure, thereby increasing the compressive force.

(32) It should be noted that FIG. **2** shows one control unit **204**, but in some other embodiments of the present application, a plurality of buttons may be included, and different buttons may be configured to perform different functions. For example, one of the buttons is used to adjust the first pressure to increase the same, and another one of the buttons is used to adjust the first pressure to

decrease the same, thereby making it easier for the user to adjust the pressure. In addition, in another embodiment, the control unit **204** may also be of another type other than a button. For example, the control unit **204** may be a key having a sliding function, and the user may control the first pressure by sliding the control unit **204**. The foregoing will not be further enumerated.
(33) In some embodiments, the pressure difference generating device **202** is further configured to, when the control unit **204** is not operated, adjust the magnitude of the first pressure to keep the compressive force stable. Such a configuration means decreases the operating difficulty of the user. For example, when the user operates the control unit **204** to control the compressive force to be at an ideal level, the user does not need to continuously maintain control of the control unit **204**, and only needs to stop operating the control unit **204** (e.g., to stop pressing the button); at which time, the pressure difference generating device **202** continuously operates to automatically adjust the magnitude of the first pressure, thereby keeping the compressive force stable.
(34) In some embodiments, the scanning assembly **200** may further include a handle **205**. The handle **205** helps the user to hold the scanning assembly **200** and position the same on the surface of the body of the person receiving a scan to perform ultrasound scanning. As shown in FIG. **2**, the handle **205** may be disposed on the housing **201**, and the control unit **204** may be disposed on the handle **205**. In such a configuration means, the user can, during ultrasound scanning, operate the control unit **204** while holding the handle **205** with two hands, thereby ensuring the continuity and stability of the scanning process. In one example, two oppositely arranged handles **205** may be included, and the user holding the same with two hands can more easily keep the scanning assembly **200** balanced.
(35) It should be noted that, although internal electrical connection structures between components are not shown in FIG. **2**, these connection means may be arbitrary ones in the prior art. For example, the components may be connected to each other by means of a cable or the like, and the cable may be disposed inside the housing **201**. The foregoing will not be described any further.
(36) After the scanning assembly **200** generates an appropriate compressive force on the tissue to be scanned, the ultrasonic transducer **203** in the scanning assembly **200** may be operated to perform ultrasound scanning on the tissue to be scanned. In some examples, the scanning assembly **200** is further provided with a driving device to drive the ultrasonic transducer **203**, thereby performing automatic scanning. Referring to FIG. **3**, FIG. **3** shows a schematic diagram of the scanning assembly **200** from which some top structures have been removed according to some embodiments of the present application.
(37) As shown in FIG. **3**, the scanning assembly **200** may include a driving device **301**. The driving device **301** is connected to the ultrasonic transducer **203** to drive the ultrasonic transducer **203** to move. Such a configuration means enables the ultrasonic transducer **203** to automatically scan the tissue to be scanned. The user does not need to hold the ultrasonic transducer **203** with their hands and move during scanning, and only needs to pay attention to the positioning of the scanning assembly **200**, thereby enhancing the degree of automation.
(38) The driving device **301** may drive, in using a variety of means, the ultrasonic transducer **203** to move. In one embodiment, the driving device **301** includes a motor **311** and a lead screw **312**. The lead screw **312** is horizontally disposed inside the housing **201**, and is connected to the ultrasonic transducer **203**. The motor **311** and the lead screw **312** cooperate with each other to drive the ultrasonic transducer **203** to move in the horizontal direction. As shown in FIG. **3**, the motor **311** is connected to one end of the ultrasonic transducer **203**, and is connected to the lead screw **312** by means of a thread. In the threaded connection means, an inner thread may be disposed on the motor **311**, and the lead screw **312** passes through the inner thread. In this way, when the motor **311** drives the inner thread to rotate, the inner thread can move in the horizontal direction relative to the lead screw **312** and drive the ultrasonic transducer **203** to move horizontally.
(39) It can be understood that the motor **311** and the lead screw **312** may cooperate in other means. For example, the motor **311** and the lead screw **312** are both fixed, and the lead screw **312** is driven

to rotate to drive the ultrasonic transducer **203** to move, or the like. Moreover, in another embodiment, the driving device may also be configured to drive the ultrasonic transducer to rotate in the housing of the scanning assembly **200**. Examples are not exhaustively enumerated.

(40) In a suitable occasion, the driving device **301** may drive the ultrasonic transducer **203** to move, so as to perform ultrasound scanning on the tissue to be scanned. In one embodiment, the above-described occasion may be controlled by the user. For example, as shown in FIG. **3**, the scanning assembly **200** may include a button **302**. The button **302** may be used to control the operation of the driving device **301**. When the button **302** is pressed by the user, the driving device **301** may continuously drive the movement of the ultrasonic transducer **203**. When the button **302** is not pressed, the driving device **301** stops operating, and the ultrasonic transducer **203** no longer moves. By adjusting the operational state of the button **302**, the user can flexibly control the movement of the ultrasonic transducer **203** according to actual scanning conditions. In another embodiment, two or more buttons **302** may be included, and different buttons may be configured to perform different functions. For example, the driving device **301** may be controlled to drive the ultrasonic transducer **203** to move in different directions. The button **302** may be positioned in a position that facilitates operation. For example, the button **302** may be positioned on the handle **205**. The button **302** may also be positioned to be opposite the control unit **204**, so that when the user holds the scanning assembly **200** with two hands, the control unit **204** may be operated with one hand, and the button **302** may be operated with the other hand, thereby enabling one operator to complete the operation. Moreover, in another embodiment, the above-described occasion may also be automatically implemented by the scanning assembly **200** (or an ultrasound imaging device). For example, when the scanning assembly **200** determines that the scanning assembly **200** is properly positioned and the compressive force for the scanning assembly is within a suitable range, the ultrasonic transducer **203** is automatically activated. A specific determination method is exemplarily described below.

(41) As described above in the present application, it is very important that the compressive force of the scanning assembly is controlled to be within a suitable range. In one aspect, the magnitude of the compressive force affects the image quality of ultrasound scanning, and in another aspect, the magnitude of the compressive force is also related to the safety of the person receiving a scan. In view of at least the above-described factors, improvements are provided in some embodiments of the present application. Referring to FIG. **4**, FIG. **4** shows a perspective bottom view of the scanning assembly **200** according to some embodiments of the present application.

(42) In some embodiments, the scanning assembly **200** further includes a film **401**. The film **401** covers the opening **211**. The compressive force can act on the tissue to be scanned (not shown in FIG. **4**) by means of the film **401**. As compared to directly pressing the housing **201** of the scanning assembly **200** onto the tissue to be scanned, the film **401** is provided to transmit the above-described compressive force, so that in one aspect, it is ensured that the compressive force is uniformly distributed, and in another aspect, the contact area of the tissue to be scanned can be increased, thereby improving safety. In one embodiment, the film **401** may be fixedly connected to the opening **211** on a lower portion of the scanning assembly **200**. In another embodiment, the film **401** can be detachably connected to the scanning assembly **200**. For example, an outer periphery of the film **401** may be fixed by means of the frame (not shown), and the film **401** may be detachably connected to the scanning assembly **200** by means of the frame. Alternatively, the film **401** may be directly detachably connected to the opening **211**. The foregoing will not be further enumerated. In actual use, the film **401** may include media commonly used in ultrasound scanning, such as a coupling medium, etc., so as to facilitate the removal of air between the tissue to be scanned and the ultrasonic transducer. The film **401** may include a transparent material to facilitate the user to observe, from above through the film **401**, the tissue to be scanned.

(43) In some embodiments, the film **401** may at least include an air-impermeable flexible film layer. The flexible film can ensure that the film **401** is closely attached to the surface of the tissue to

be scanned, thereby preventing air leakage. The air-impermeable flexible film can ensure that when a negative pressure is formed between the tissue to be scanned and the scanning assembly, the negative pressure directly acts on the air-impermeable flexible film, and the tissue to be scanned is tightly sucked by means of the air-impermeable flexible film. As compared to directly tightly sucking the tissue to be scanned, the air-impermeable film can further improve the degree of comfort and the scanning safety of the person receiving a scan. It can be understood that in some examples, the film **401** may include a plurality of layers of composite material. For example, the film **401** may include an air-permeable film and an air-impermeable film. The air-impermeable film is used to improve the degree of comfort of the person receiving a scan, and the air-permeable film is for penetration of the coupling medium.

(44) To further improve the sealing between the scanning assembly **200** and the tissue to be scanned, the scanning assembly **200** in some embodiments of the present application further includes a seal ring **402**. The seal ring **402** is made of a flexible material, and is disposed around the opening **211**. The flexible material can ensure good contact between the seal ring **402** and the tissue to be scanned so as to prevent air leakage, and may specifically be of an arbitrary material type in the art, such as silicone, rubber, or the like. Details will not be described herein again. In addition, the seal ring **402** can further improve the degree of comfort of the person receiving a scan. In some examples, the seal ring **402** is configured to be connected around the opening **211**. In some other examples, the seal ring **402** may also be configured to be integral with the film **401**, so that when cleaning and sterilizing or replacement is needed, the seal ring **402** and the film **401** can be removed from the scanning assembly **200** simultaneously.

(45) The scanning assembly **200** of the present application may further include a compressive force measurement unit. The compressive force measurement unit may be used to measure the compressive force of the scanning assembly **200** for the tissue to be scanned. Such a configuration means helps the compressive force borne by the tissue to be scanned to be accurately acquired in a scanning process, so that the scanning assembly **200** can control the pressure difference generating device of any embodiment described above to adjust the first pressure, thereby dynamically adjusting the compressive force.

(46) In some embodiments, the compressive force measurement unit may include an air pressure sensor. The air pressure sensor is configured to be inside the housing **201** of the scanning assembly **200**, so as to measure in real time the air pressure difference between the interior and the exterior of the housing **201**. Further, the magnitude of the compressive force can be calculated according to the air pressure difference and the weight of the scanning assembly **200**.

(47) In some other embodiments, the compressive force measurement unit may, as shown in FIG. **4**, include at least one force sensor **403**. The force sensors **403** are disposed below the housing **201**. The force sensors **403** can directly measure the compressive force acting on the surface of the tissue to be scanned. In one example, one force sensor **403** is provided. In another example, a plurality of force sensors **403** are provided and are uniformly distributed below the housing **201**, a plurality of force sensors **403** enable more comprehensive and accurate measurements of the compressive force. Moreover, the difference between values of the plurality of force sensors **403** can further be used to determine whether the compressive force of the scanning assembly **200** on the tissue to be scanned is uniformly applied.

(48) A method of using the scanning assembly disclosed in any embodiment described above is exemplarily described below.

(49) In one example, the scanning assembly **200** described above can automatically regulate the compressive force. Specifically, the scanning assembly **200** may measure, in real time by means of the above-described compressive force measurement unit, the compressive force of the scanning assembly **200** acting on the tissue to be scanned. When a predetermined compressive force is reached, the pressure difference generating device can dynamically adjust the air pressure difference between the interior and the exterior of the scanning assembly **200** to adjust the first

pressure, thereby achieving the objective of regulating the current compressive force. If the compressive force measurement unit detects that the current compressive force is smaller than the predetermined compressive force, the pressure difference generating device increases the air pressure difference to increase the compressive force. If the compressive force measurement unit detects that the current compressive force is greater than the predetermined compressive force, the pressure difference generating device automatically decreases the air pressure difference to decrease the compressive force. In this way, the compressive force is automatically maintained at a stable level, thereby facilitating ultrasound scanning. The predetermined compressive force may be configured by the user by operating the control unit as described above in the present application, or may be preset by the ultrasound imaging device.

(50) In one example, the scanning assembly **200** described above may further be used to further ensure the safety of the person receiving a scan. For example, an upper limit of the compressive force may be preset. Once the compressive force measurement unit detects that the compressive force reaches the upper limit, the scanning assembly **200** may stop the pressure difference generating device operating, thereby rapidly decreasing the compressive force.

(51) In one example, the scanning assembly **200** described above can further automatically scan the site to be scanned. For example, when the user correctly positions the scanning assembly **200** and activates the pressure difference generating device, the pressure difference generating device starts to gradually increase the air pressure difference to increase the compressive force. The process of increasing described above may be operated by the user, or may be automatically controlled by the pressure difference generating device. In this case, the compressive force measurement unit measures in real time the current compressive force. Once the compressive force reaches a predetermined value, in one aspect, the scanning assembly **200** may dynamically adjust the pressure difference generating device to cause the current compressive force to remain substantially stable, and in another aspect, the scanning assembly **200** may further automatically activate the ultrasonic transducer to perform scanning. In such an embodiment, the components cooperate with each other, thereby greatly enhancing the degree of automation and simplifying workflow.

(52) It can be understood that the above merely exemplarily describes an operating means of the scanning assembly **200**, and the actual operating means of the scanning assembly **200** may also be of other types. In addition, the scanning assembly **200** may further include other components, e.g., a processor circuit for cooperating with the component described in any embodiment described above to perform control. The processor circuit may be integrated in the scanning assembly **200**, or may be integrated in the ultrasound imaging device connected to the scanning assembly **200**. An ultrasound imaging device connected to the scanning assembly of any embodiment of the present application is exemplarily described below.

(53) Referring to FIG. **5**, FIG. **5** shows a schematic diagram of an ultrasound imaging device **500** including a scanning assembly according to some embodiments of the present application. The ultrasound imaging device **500** may include the scanning assembly described above in any described embodiment of the present application.

(54) In the embodiments described in FIG. **5**, the scanning assembly **200** may further include a cable **501** and a connector **502**. One end of the cable **501** is connected to the scanning assembly **200**. The connector **502** is connected to the other end of the cable **501**. The scanning assembly **200** is electrically connected to the ultrasound imaging device **500** by means of the cable **501** and the connector **502**, so that power and data transmission are performed by using the cable **501**.

(55) Such a configuration means ensures that the scanning assembly **200** is flexibly used. In particular, as compared to the existing ultrasound imaging device described above, the scanning assembly **200** no longer needs to be connected to the ultrasound imaging device by means of components including an adjustable arm, and is connected thereto by means of only the flexible cable. In addition, the pressure difference generating device described above is directly integrated on the scanning assembly **200**, and likewise does not affect the flexibility of use of the scanning

assembly **200**. As shown in FIG. **5**, an operator **503** can flexibly use the scanning assembly **200** to position a site to be scanned on a person **504** receiving a scan.

(56) It should be noted that the connector **502** may be any type of connector in the prior art, and may be configured to be detachably connected to the ultrasound imaging device **500**. For example, the ultrasound imaging device **500** may be a general-purpose ultrasound imaging device, and is adapted to be connected to different types of probes to scan different sites to be scanned. Such an ultrasound imaging device **500** typically has several probe connector sockets. Correspondingly, the connector **502** may be configured to match at least one socket. In which case, the scanning assembly **200** no longer needs to be configured with the dedicated ultrasound imaging device **500**, and the scanning assembly **200** may be used as one of the ultrasound probes. Therefore, the versatility of the ultrasound imaging device **500** can be significantly improved. In another example, the connector may further be fixedly connected to the ultrasound imaging device **500**, and details will not be described again.

(57) The above-described scanning assembly **200** is an example of the present application. In another example, the cable and the connector may be removed. For example, the scanning assembly **200** may perform ultrasound scanning at a higher degree of freedom by means of modules such as an internal power source, the processor circuit, etc. The scanning assembly **200** may further have a built-in wireless signal receiving/sending module etc., to perform data transmission. In addition, the scanning assembly **200** may further be configured with a memory to store data and export the same by using a suitable means (e.g., to export the same in a wired or wireless means).

(58) With continued reference to FIG. **5**, other components of the ultrasound imaging device **500** are exemplarily described. In addition to the scanning assembly **200** of any embodiment described above, the ultrasound imaging device **500** may further include a main body **505** and a display **506**.

(59) The display **506** may be configured to display ultrasound image information acquired by means of ultrasound scanning, to allow the user to view the same. In another example, information related to parameters of scanning (such as the progress of scanning) may be sent to the display **506** so as to be displayed. The display **506** may include a user interface configured to display images or other information to the user. The type of the display **506** may be diverse, such as an LCD or OLED screen, or the like. In one example, the display **506** may further be configured to be a touch screen to facilitate the user in perform inputting.

(60) The main body **505** may include a processor (not shown in the drawing). The processor (and a circuit thereof) may be used to control assemblies including the scanning assembly **200** so as to perform ultrasound scanning and process an echo signal to acquire an ultrasound image, and can control the display **506** to display the ultrasound image. In one example, the scanning assembly **200** described above in the present application performs an ultrasound scanning process independently. In another example, the above-described scanning process may be completed under the control of the processor in the main body **505**. That is, when the scanning assembly **200** is electrically connected to the ultrasound imaging device **500**, the processor can communicate with the scanning assembly **200**, thereby sending/receiving data and controlling the scanning assembly **200**. For example, the processor may be used to process data acquired by the ultrasonic transducer in the scanning assembly **200** by means of ultrasound scanning, so as to generate an ultrasound image. Further, the processor may further be configured to control the pressure difference generating device of the scanning assembly **200** on the basis of the ultrasound image, so as to adjust the first pressure. For example, as an alternative to the solution described in the above embodiments, the processor may determine quality of the ultrasound image by means of an algorithm, so as to automatically control the pressure difference generating device device.

(61) The main body **505** may further include a memory (not shown in the drawing). The memory may include movable and/or permanent apparatuses, and may include an optical memory, a semiconductor memory, and/or a magnetic memory, etc. The memory may include a volatile, non-

volatile, dynamic, static, read/write, read only, random access, sequential access, and/or additional memory. The memory may store non-transitory instructions executable by the processor. The memory may further store raw image data received from the scanning assembly **200**, a processed ultrasound image received from the processor, etc.

(62) It can be understood that the above merely exemplarily describes the ultrasound imaging device **500**. The ultrasound imaging device **500** may further be configured to include other parts, such as a user input device for facilitating human-machine interaction. The user input device may be the touch screen described above, or may be a mouse, a keyboard, or the like. The foregoing will not be further enumerated.

(63) The purpose of providing the above specific embodiments is to allow the disclosure of the present application to be understood more thoroughly and comprehensively; however, the present application is not limited to said specific embodiments. A person skilled in the art should understand that various modifications, equivalent replacements, changes and the like can be further made to the present application and should be included in the scope of protection of the present application as long as these changes do not depart from the spirit of the present application.

Claims

1. A scanning assembly, comprising: a housing, wherein a lower portion of the housing comprises an opening, and the opening being configured to be covered by tissue to be scanned; a pressure difference generating device disposed on the housing and used to generate an air pressure difference between the interior and the exterior of the housing, the air pressure difference providing a first pressure for the tissue to be scanned, the weight of the scanning assembly providing a second pressure for the tissue to be scanned, the tissue to be scanned being compressed under the action of a compressive force, and the compressive force comprising the first pressure and the second pressure; and an ultrasonic transducer disposed in the housing and used to perform ultrasound scanning on the compressed tissue to be scanned.
2. The scanning assembly according to claim 1, further comprising: a film covering the opening, the compressive force acting on the tissue to be scanned by means of the film.
3. The scanning assembly according to claim 2, wherein the film at least comprises an air-impermeable flexible film layer.
4. The scanning assembly according to claim 1, further comprising: a driving device connected to the ultrasonic transducer to drive the ultrasonic transducer to move.
5. The scanning assembly according to claim 4, wherein the driving device comprises an electric motor and a lead screw, the lead screw being horizontally disposed in the housing and being connected to the ultrasonic transducer, and the electric motor and the lead screw cooperating with each other to drive the ultrasonic transducer to move in the horizontal direction.
6. The scanning assembly according to claim 1, wherein the pressure difference generating device is configured to adjust the first pressure by adjusting the magnitude of the air pressure difference.
7. The scanning assembly according to claim 6, wherein the pressure difference generating device comprises a fan, and the rotational speed of the fan is configured to be adjustable to adjust the air pressure difference.
8. The scanning assembly according to claim 6, wherein the pressure difference generating device is further configured to automatically adjust the first pressure to keep the compressive force stable.
9. The scanning assembly according to claim 6, further comprising: a control unit configured to, when operated, control the pressure difference generating device adjust the magnitude of the compressive force by adjusting the magnitude of the first pressure.
10. The scanning assembly according to claim 9, wherein the pressure difference generating device is further configured to, when the control unit is not operated, keep the compressive force stable by adjusting the magnitude of the first pressure.

11. The scanning assembly according to claim 9, further comprising: a handle disposed on the housing, the control unit being disposed on the handle.
12. The scanning assembly according to claim 1, further comprising: a compressive force measurement unit for measuring the compressive force.
13. The scanning assembly according to claim 12, wherein the compressive force measurement unit comprises at least one force sensor, and the at least one force sensor is disposed below the housing.
14. The scanning assembly according to claim 1, further comprising: a seal ring made of a flexible material and disposed around the opening.
15. The scanning assembly according to claim 1, further comprising: a cable, one end of the cable being connected to the scanning assembly; and a connector which is connected to the other end of the cable, the scanning assembly being electrically connected to an ultrasound imaging device by means of the cable and the connector, so that power and data transmission are performed by using the cable.
16. An ultrasound imaging device, comprising the scanning assembly according to claim 1.
17. The ultrasound imaging device according to claim 16, further comprising: a processor configured to process data acquired by the ultrasonic transducer by means of ultrasound scanning to generate an ultrasound image, the processor being further configured to, on the basis of the ultrasound image, control the pressure difference generating device to adjust the first pressure.